

Hamiltonian-Geometric Optimization Framework for Machine Learning

A Concise Walkthrough: Framework, Validation, Evidence

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Capacitor Fringes Project

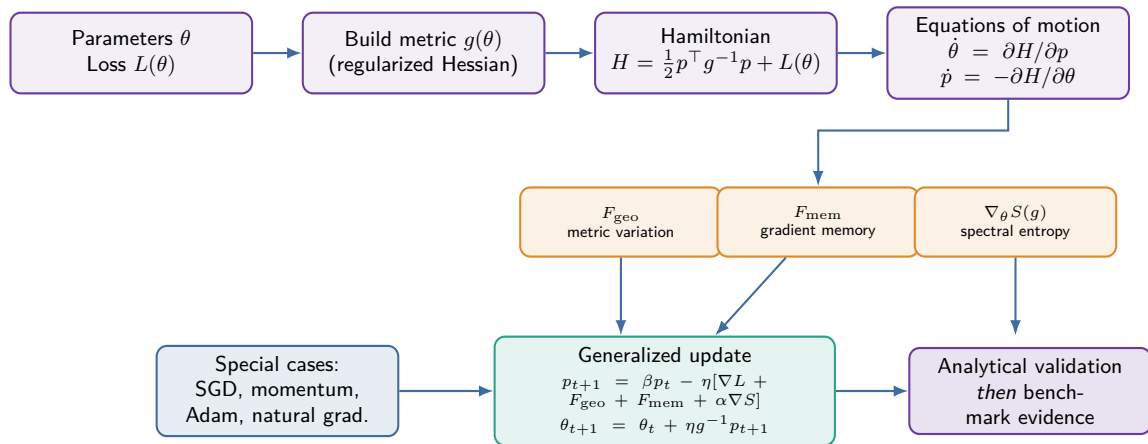
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Goal

Can a single **Hamiltonian-geometric** update rule act as a useful optimizer family across classical ML, LLM-style training, PINNs, and quantum-control objectives?

- Treat parameters θ as generalized coordinates, the loss $L(\theta)$ as potential energy, and optimizer state as momentum p evolving on a **metric manifold** $g(\theta)$.
- One family should recover SGD, momentum, Adam, and natural gradient as special cases.
- Claim is deliberately *bounded*: not uniformly superior, strongest when full-metric curvature is exploitable.

Methodology — Mathematical Framework at a Glance



Deterministic stages left to right; forces merge into one update rule; output feeds validation before benchmarking.

$$H(\theta, p) = \frac{1}{2} p_i g^{ij}(\theta) p_j + L(\theta),$$
$$\dot{\theta}^i = \frac{\partial H}{\partial p_i} = g^{ij}(\theta) p_j, \quad \dot{p}_i = -\frac{\partial H}{\partial \theta^i} = -\partial_i L - \frac{1}{2} p_j p_k \partial_i g^{jk}(\theta).$$

- Second term in $\dot{p}_i$ is the **geometric force** from a coordinate-dependent metric.
- Dissipation is added so dynamics *optimize* rather than conserve energy.
- Same structure carries the memory and spectral-entropy correction terms.

One Update Rule, Many Optimizers

Optimizer	Hamiltonian-geometric specialization
SGD / heavy-ball	Flat metric, $F_{\text{geo}} = 0$, instant or finite momentum relax
Adam / RMSProp	Diagonal adaptive metric; ignores F_{geo} from metric variation
Natural gradient	Full information metric, usually no momentum or memory force
Hamiltonian-geometric	Full metric <i>with</i> momentum, memory, spectral entropy, curvature

- Modern optimizers (Shampoo, K-FAC, Muon, AdaFactor, LARS/LAMB) sit at intermediate points on the same metric-structure hierarchy — from full $n \times n$ down to a single global scalar.

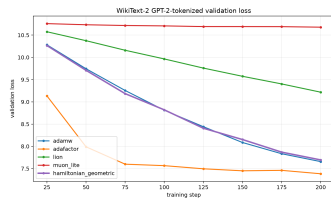
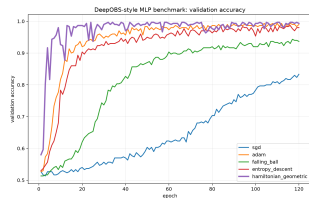
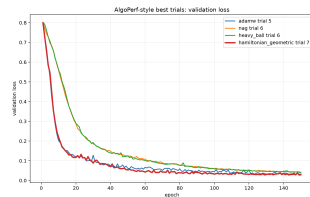
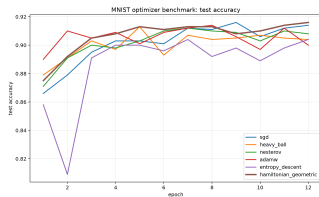
Analytical Validation (Equation-Level, All PASS)

What & reference: each row checks one equation from Sparsho Chakraborty's *literature-review PDF* — the design reference for this optimizer's architecture — against a finite-difference computation on a fixed, seeded, genuinely curved-metric toy problem.

Check	Kind	Max error	Tol.	Result
Legendre transform	literal	2.96×10^{-11}	10^{-6}	PASS
Geometric force	literal	4.60×10^{-11}	10^{-5}	PASS
Flat-metric reduction	literal	0.000	10^{-9}	PASS
Memory recursion	literal	8.88×10^{-16}	10^{-9}	PASS
Hessian regularization	counterex.	0.000	10^{-9}	PASS
Rayleigh dissipation (curved metric)	completed	1.64×10^{-9}	10^{-6}	PASS
Spectral entropy gradient	completed	1.14×10^{-10}	10^{-5}	PASS

- **literal** = tests a PDF equation exactly as printed; **completed** = tests our documented completion of a formula the PDF states but never fully derives; **counterex.** = shows the PDF's literal condition fails, then confirms the shipped code fix succeeds.

Benchmark Evidence Snapshot



Benchmark Results — Classical, PINN & LLM

Workload	Hamiltonian-geometric	Best other optimizer	Reading
MNIST softmax (test)	Loss 0.2938, acc. 0.9160	Nesterov: loss 0.2962	Best
AlgoPerf-style MLP (val.)	Loss 0.0347, acc. 0.9867	AdamW: loss 0.0430	Best
DeepOBS-style MLP (val.)	Loss 0.0203, acc. 0.9933	Adam: loss 0.0449	Best
Capacitor PINN (final)	Loss 17.13218	Entropy descent: 17.31064	Best
WikiText-2 LLM (val., 200 steps)	Loss 6.778, ppl. 878.3	AdaFactor: loss 6.765, ppl. 866.8	2nd best, fastest wall-clock

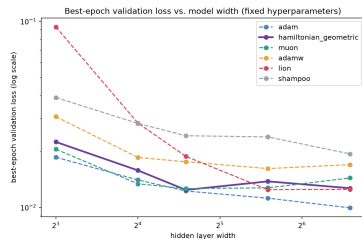
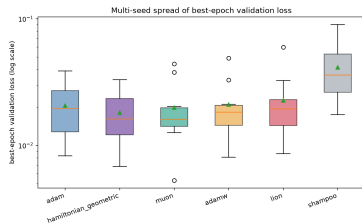
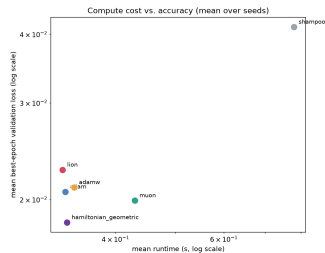
- On the LLM run, AdaFactor edges it out on loss, but Hamiltonian-geometric clearly beats AdamW (6.843) and Lion (7.032), and finishes fastest of the five; Muon-lite diverges (loss 9.31) under this untuned configuration.
- Small-scale, single-seed runs throughout — evidence of mechanism, not a scaling proof.

Multi-Seed Reality Check — Do the Single-Seed Wins Hold Up?

Workload	n	HG mean $\pm$ SD	Best-other mean $\pm$ SD	Paired p
MNIST (test loss)	10	0.336 ± 0.027	0.338 ± 0.028	0.688
AlgoPerf-style (val. loss)	10	0.0485 ± 0.028	0.0549 ± 0.026	0.067
DeepOBS-style (val. loss)	10	0.0580 ± 0.039	0.0551 ± 0.025	0.689
WikiText-2 LLM (val. loss)	3	7.368 ± 0.015	7.237 ± 0.006	0.003

- **DeepOBS-style is the most consequential reversal here:** the single-seed table showed a $2\times$ win; across 10 seeds HG wins only 5/10 — a coin flip, $p = 0.689$. That single-seed entry was a favorable draw of the seed, not a reliable advantage.
- MNIST: HG wins 7/10 seeds but not significantly ($p = 0.688$). AlgoPerf-style: borderline favorable (8/10, $p = 0.067$).
- **WikiText-2 LLM: the opposite correction.** AdaFactor beats HG in all 3 seeds, and the gap is significant ($p = 0.003$) despite only 3 seeds — the earlier “2nd best” framing understated how consistent AdaFactor’s edge actually is.
- PINN multi-seed was attempted but not completed: its finite-difference metric makes a single seed far more expensive than the other three combined — an honest, costed gap, not a result.

Modern Optimizer Comparison



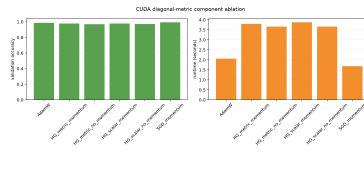
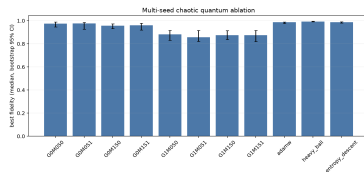
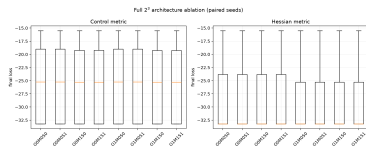
Adam, Hamiltonian-geometric (diagonal-metric reduction), and Muon lead the pack; the 10-seed sweep (center) holds the near-tie, but the width-scaling sweep (right) favors Adam — competitive evidence, not decisive dominance.

Benchmark Results — Modern Optimizers (Spiral MLP)

Optimizer	Best epoch	Val. loss	Val. acc.
Adam	102	0.012244	0.9967
Hamiltonian-geometric	120	0.012416	0.9967
Muon	102	0.012624	0.9967
AdamW	91	0.017588	0.9967
Lion	113	0.018797	0.9967
Shampoo	117	0.024266	0.9933

- Ranked by best-epoch validation loss, lower is better.
- Adam, Hamiltonian-geometric, and Muon differ by at most 4×10^{-4} in absolute validation loss — a narrow leading group, not a clearly separated ordering.
- AdamW and Lion trail by a larger margin; Shampoo is both slower and slightly worse here.

Extended Ablation Study — Design & Evidence



A separate, more rigorous 2^3 factorial (geometric $\times$ memory $\times$ spectral force), paired-seed replication: 40 seeds/300 steps (architecture, left), 10 seeds/40 iterations (chaotic quantum, center), 10 seeds/600 steps (CUDA diagonal reduction, right).

Extended Ablation Study — Key Numbers

Study	Factor	Effect	p
Architecture, control metric	G, M, S	≤ 0.59	all > 0.2 (n.s.)
Architecture, Hessian metric	S (spectral)	2.08	9.25×10^{-5}
Architecture, Hessian metric	G, M	≤ 0.32	> 0.5 (n.s.)
Quantum (kicked-top)	G (geometric)	0.88	1.57×10^{-4}

- **Architecture (40 seeds):** confirms the earlier story — the spectral term only matters when the metric is genuinely tied to loss curvature (Hessian), not under an unrelated control metric.
- **Quantum (10 seeds, paired):** here the honest result is negative — turning on the geometric force *significantly lowers* fidelity (0.86–0.88 vs. 0.96–0.98 with it off), and every simple baseline (heavy-ball 0.993, entropy descent 0.988, AdamW 0.986) beats every full-HG variant, while costing 10–20 $\times$ more runtime.
- **CUDA scaling (10 seeds):** the diagonal-metric Torch reduction (0.978 acc.) trails plain SGD (0.993) and AdamW (0.987) here too, at roughly 2 $\times$ the runtime.
- **Rigor rewards honesty:** this replication does not uniformly favor Hamiltonian-geometric — it sharpens exactly where the framework helps (curvature-tied spectral term) and where it currently does not (quantum control, GPU-scale diagonal reduction).

Limitations

- **Scale:** all systems are small; full $n \times n$ metric is mechanism evidence, not a production-scale replacement for Adam-family optimizers.
- **Statistics:** multi-seed variance now covers the modern-optimizer sweep (10 seeds), the classical/LLM reality check (10/10/10/3 seeds), and the extended ablation (40/10/10 seeds) — and it is not uniformly favorable: the DeepOBS single-seed win does not replicate, and the LLM loss to AdaFactor is significant, not a near-tie.
- **Tuning:** learning rates from modest grids, not exhaustive search.
- **Comparability:** the modern-optimizer and CUDA-ablation runs both use a diagonal reduction, not the full curvature-memory-entropy architecture.

Summary

- Formulation is mathematically coherent once the spectral-force sign convention and memory-metric construction are made explicit.
- Evidence is strongest where exploitable full-metric geometry is present.
- Places the method in the leading group of modern optimizers, without separating it decisively from Adam or Muon.
- The rigorous extended ablation confirms this precisely: the spectral term earns its keep only under a genuine Hessian metric, while quantum and GPU-scale results caution against over-claiming.
- The multi-seed reality check cuts both ways too — it takes away a single-seed win (DeepOBS) as much as it takes away worry about a near-tie (LLM loss to AdaFactor is real).

Thank You

Hamiltonian-Geometric Optimization Framework